NEW YORK STATE CRIME PROFILING

IA650 - DATA MINING

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INTRODUCTION

Crime Data Analysis:

- Helps identify patterns and differences across regions, agencies, and types of crimes.
- It becomes possible to understand how crime varies from one county to another and which areas show similar behavior.

BACKGROUND

- The dataset contains crime records reported by agencies across New York counties.
- Data from 1990 to 2024.
- It includes both violent and property crime variables.

OBJECTIVE

The objective of this project is to:

- To cluster New York State counties based on crime patterns.
- To predict the murder rate using other crime-related features.

SAMPLE DATA

SOURCE:https://data.ny.gov/Public-Safety/Index-Crimes-by-County-and-Agency-Beginning-1990/ca8h-8gjq/

https://data.ny.gov/Government-Finance/Annual-Population-Estimates-for-New-York-State-and/krt9-ym2k/about_data

County	Agency	Year	Months Reported	Index Total	Violent Total	Murder	Rape	Robbery	Aggravated Assault	Property Total	Burglary	Larceny	Motor Vehicle Theft	Region
Albany	Albany City PD	2024	12	4057	985	13	90	233	649	3072	409	2360	303	Non-New York City
Albany	Albany County Park PD	2024	12	0	0	0	0	0	0	0	0	0	0	Non-New York City
Albany	Albany County Sheriff	2024	12	58	8	0	1	1	6	50	7	40	3	Non-New York City
Albany	Albany County State Police	2024	11	181	58	1	12	1	44	123	4	112	7	Non-New York City
Albany	Altamont Vg PD	2024	12	5	0	0	0	0	0	5	0	5	0	Non-New York City

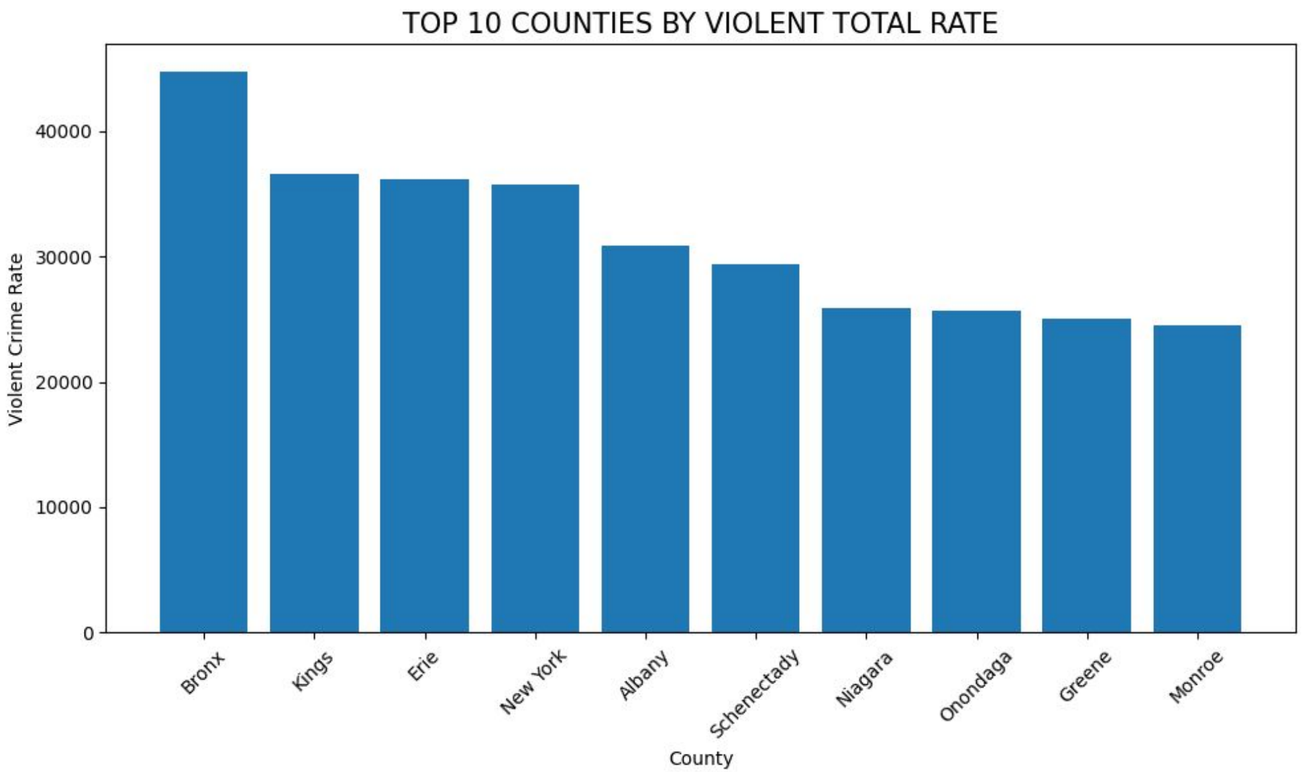
DATA PRE PROCESSING

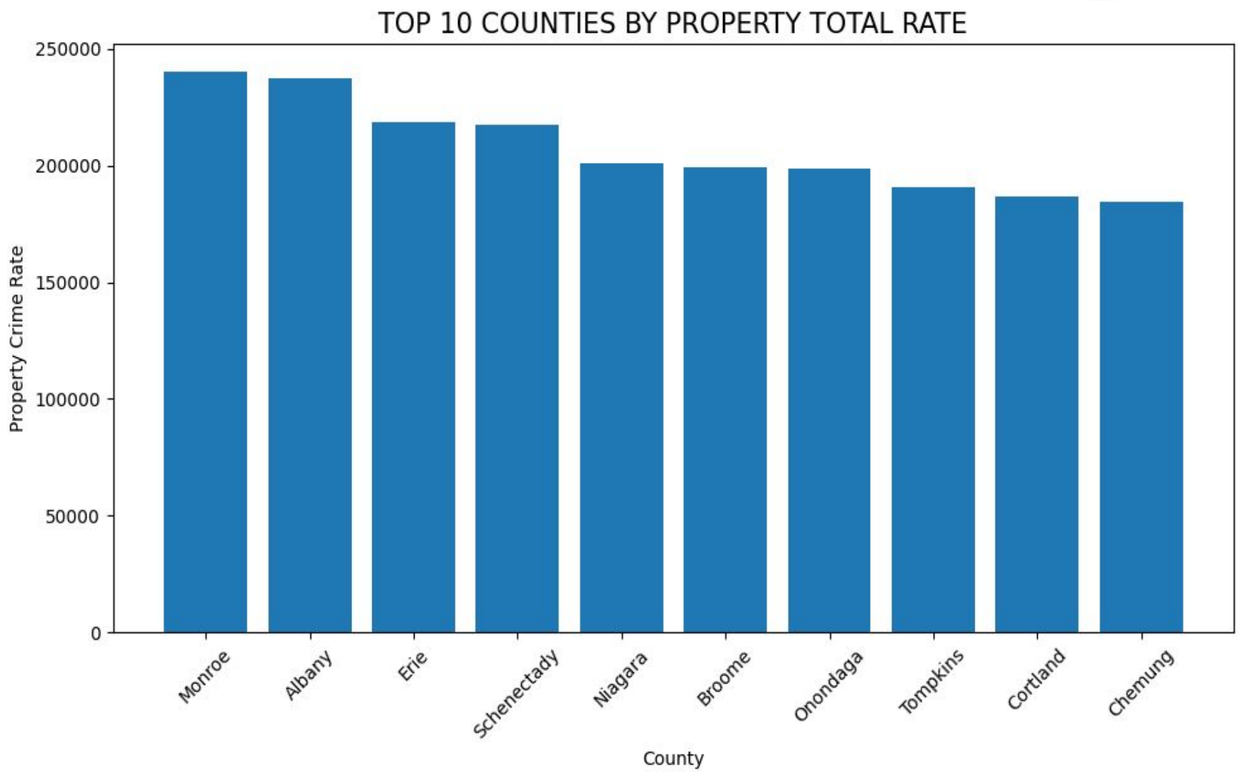
- Removed duplicate values.
- Aggregated crime data at the **county-year level** by summing all agency reported within each county for a given year.
- Merged the crime dataset with the population dataset using **County** and **Year**.
- Added the **Population** column after merging.
- Created population-adjusted variables: **Index Total Rate, Violent Total Rate, Property Total Rate etc.**
- This helped standardize crime values and made county-level comparison fairer

DATASET DESCRIPTION

No.	Variable Name	Type	Description
1	County	Text	County or location where the crime was reported.
2	Year	Integer	Year in which the crime incident was reported.
3	Index total	Integer	Total number of reported index crimes, including both violent and property crimes.
4	Violent Total	Integer	Total number of reported violent crimes.
5	Murder	Integer	Number of reported murder victims or murder cases.
6	Rape	Integer	Number of reported forcible rape victims or incidents.
7	Robbery	Integer	Number of reported robbery incidents.
8	Aggravated Assault	Integer	Number of reported aggravated assault incidents.
9	Property Total	Integer	Total number of reported property crimes.
10	Burglary	Integer	Number of reported burglary incidents.
11	Larceny	Integer	Number of reported larceny incidents.
12	Motor Vehicle Theft	Integer	Number of reported motor vehicle theft incidents.
13	Population	Integer	Total population of the county in the given year, used to calculate standardized crime rates
14	Total_Index_Rate	Float	Total index crimes per 100,000 population.
15	Violent_Crime_Rate	Float	Total violent crimes per 100,000 population.
16	Murder_Rate	Float	Number of murders per 100,000 population.
17	Rape_Rate	Float	Number of forcible rapes per 100,000 population.
18	Robbery_Rate	Float	Number of robberies per 100,000 population.
19	Aggravated_Assault_Rate	Float	Number of aggravated assaults per 100,000 population.
20	Property_Crime_Rate	Float	Total property crimes per 100,000 population.
21	Burglary_Rate	Float	Number of burglaries per 100,000 population.
22	Larceny_Rate	Float	Number of larcenies per 100,000 population.
23	Motor_Vehicle_Theft_Rate	Float	Number of motor vehicle thefts per 100,000 population.

OVERALL CRIME COMPARISON

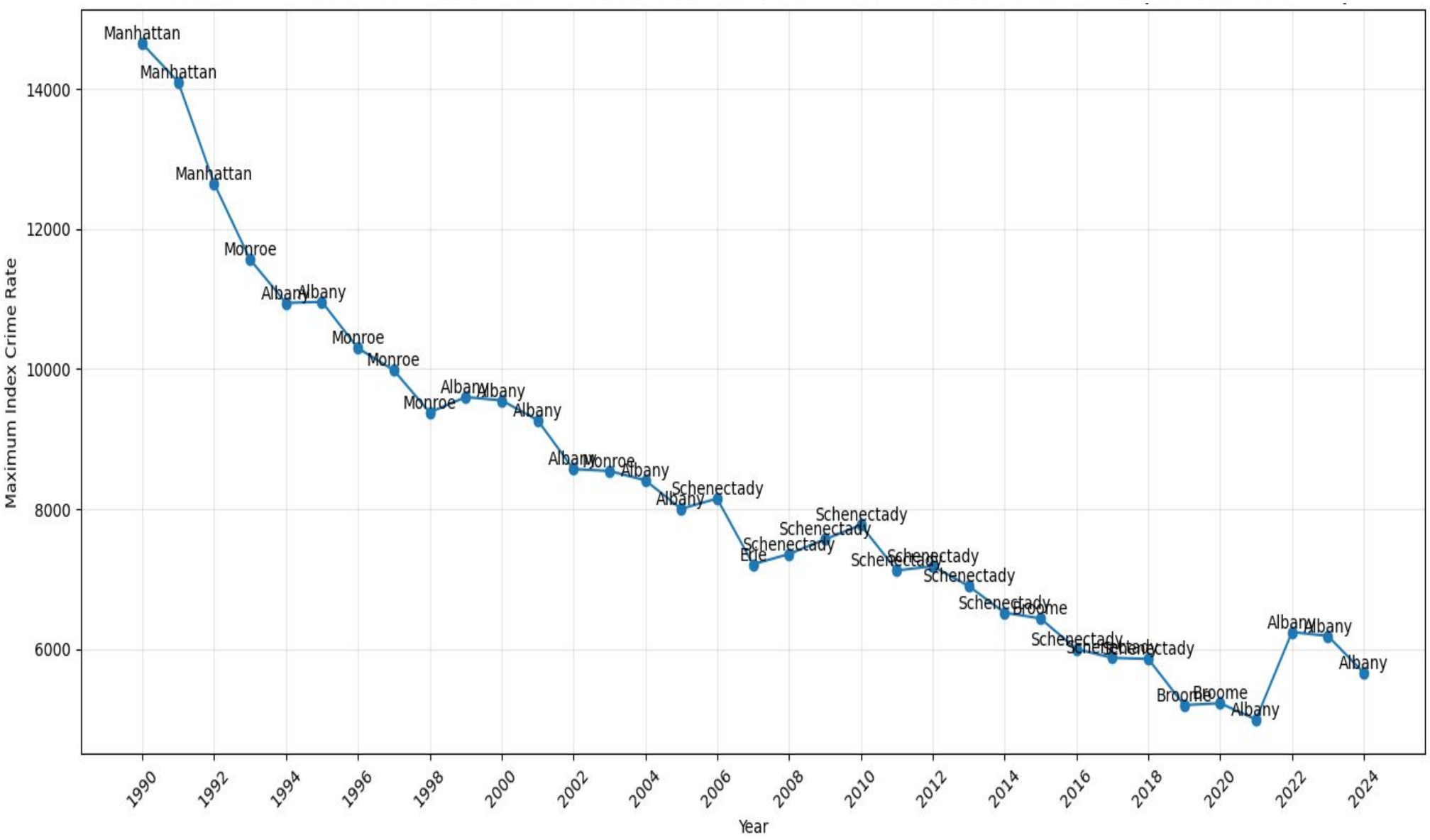




- Bronx shows the strongest violent crime burden among the counties shown.
- Higher violent crime rates are more visible in a few major urban counties.

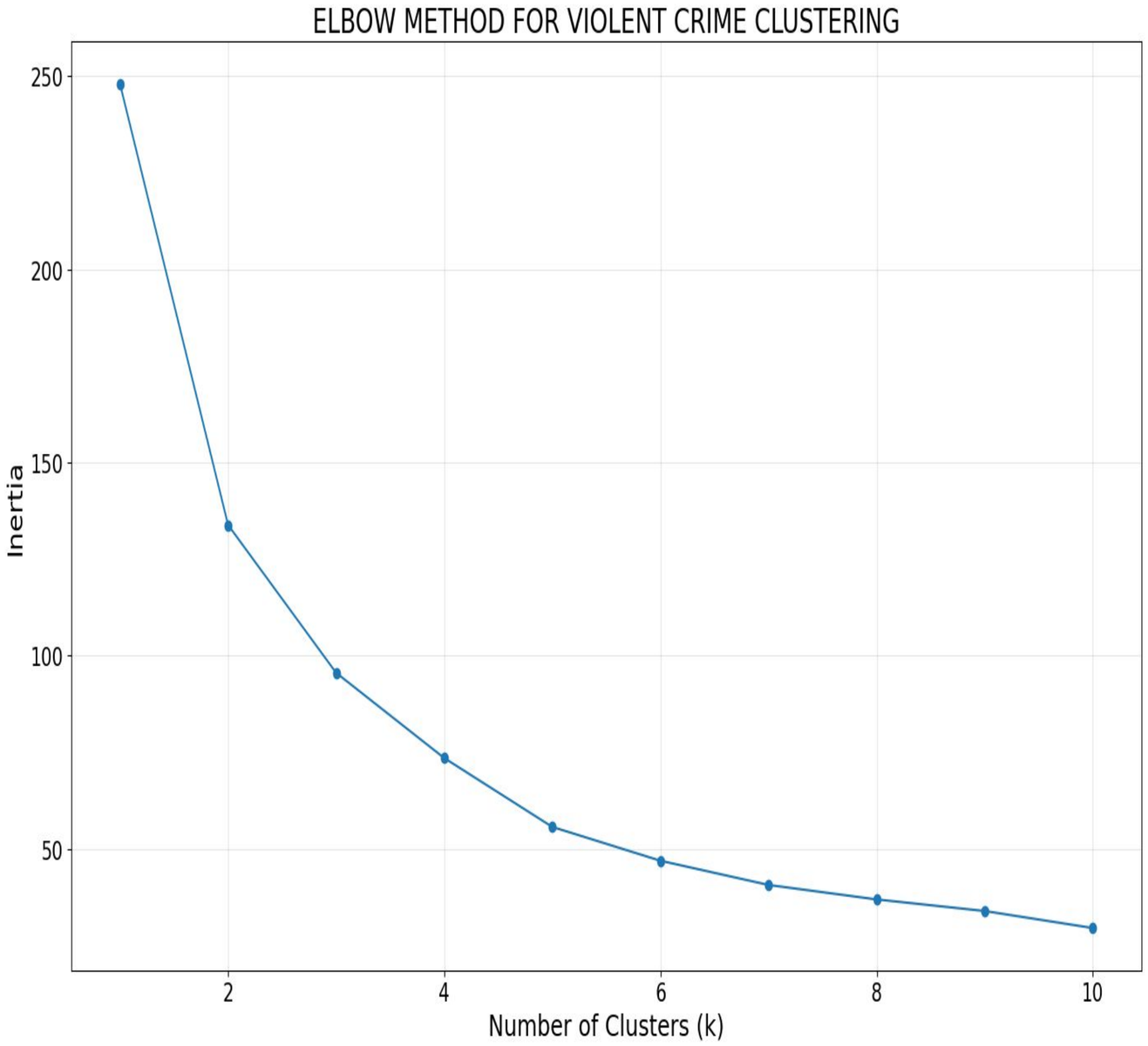
- Monroe leads in property crime rate, followed by Albany and Erie.
- Property crime appears more dispersed across counties than violent crime.

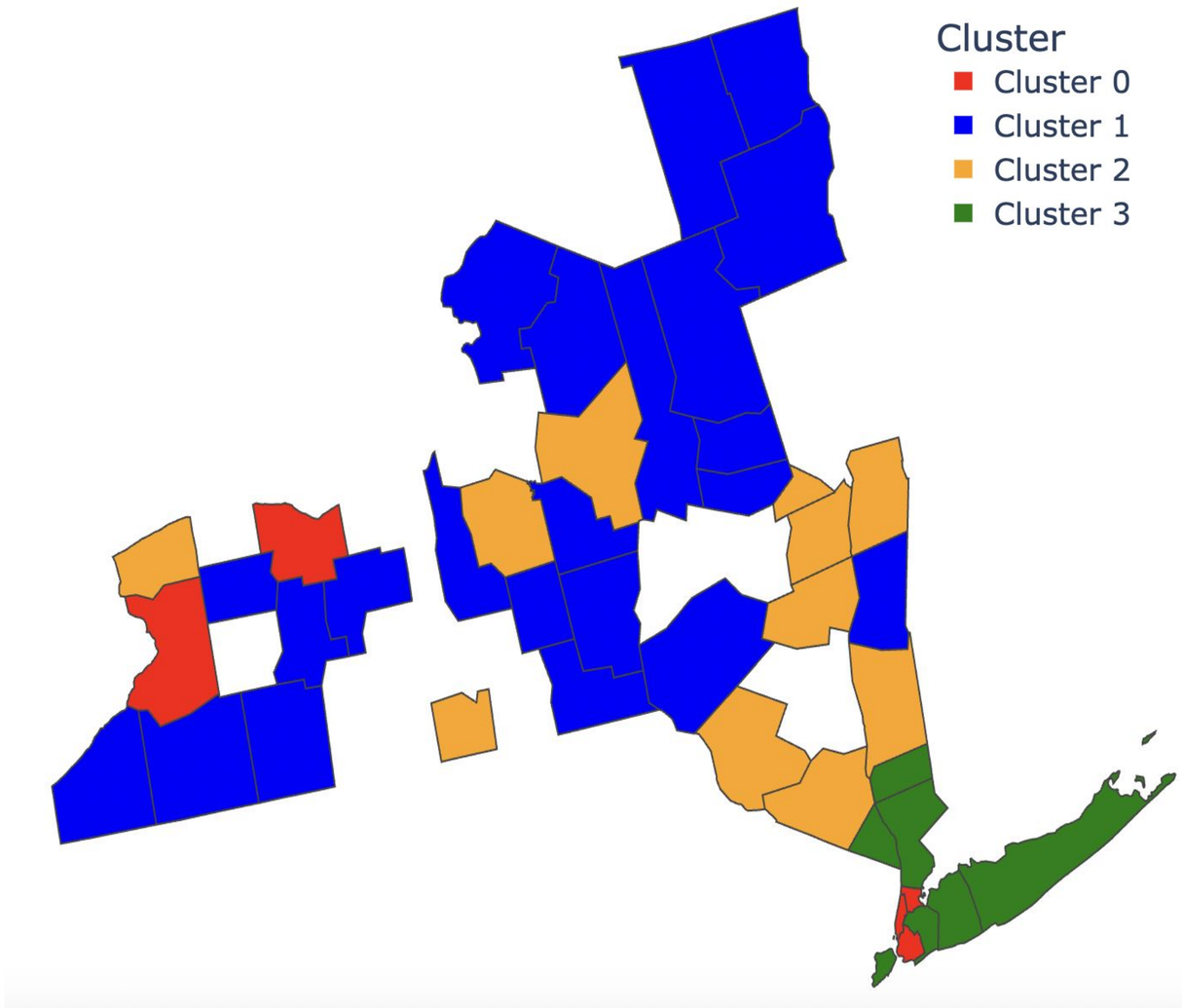
COUNTY WITH MAXIMUM INDEX CRIME RATE EACH YEAR (1990-2024)



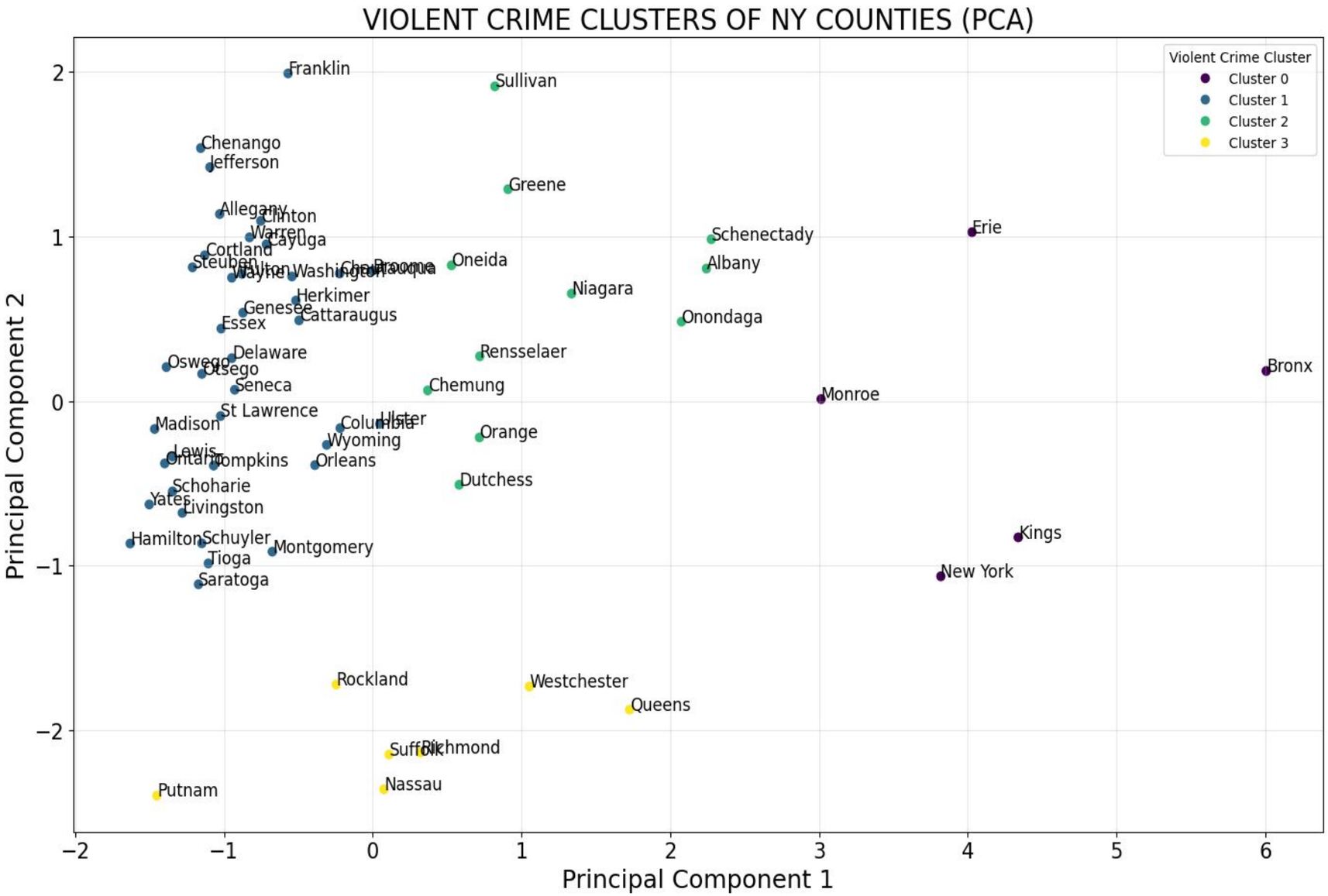
- Shows overall decline from 1990 to 2024.
- Manhattan dominated the early years, but later the highest rate shifted to other counties such as Albany and Schenectady.
- This suggests that the location of the highest crime burden changed over time, even as the overall peak rate decreased.

CLUSTERING


ELBOW METHOD FOR VIOLENT CRIME CLUSTERING
Inertia
250
200
150
100
50
2
4
6
8
10
Number of Clusters (k)


Cluster
Cluster 0
Cluster 1
Cluster 2
Cluster 3

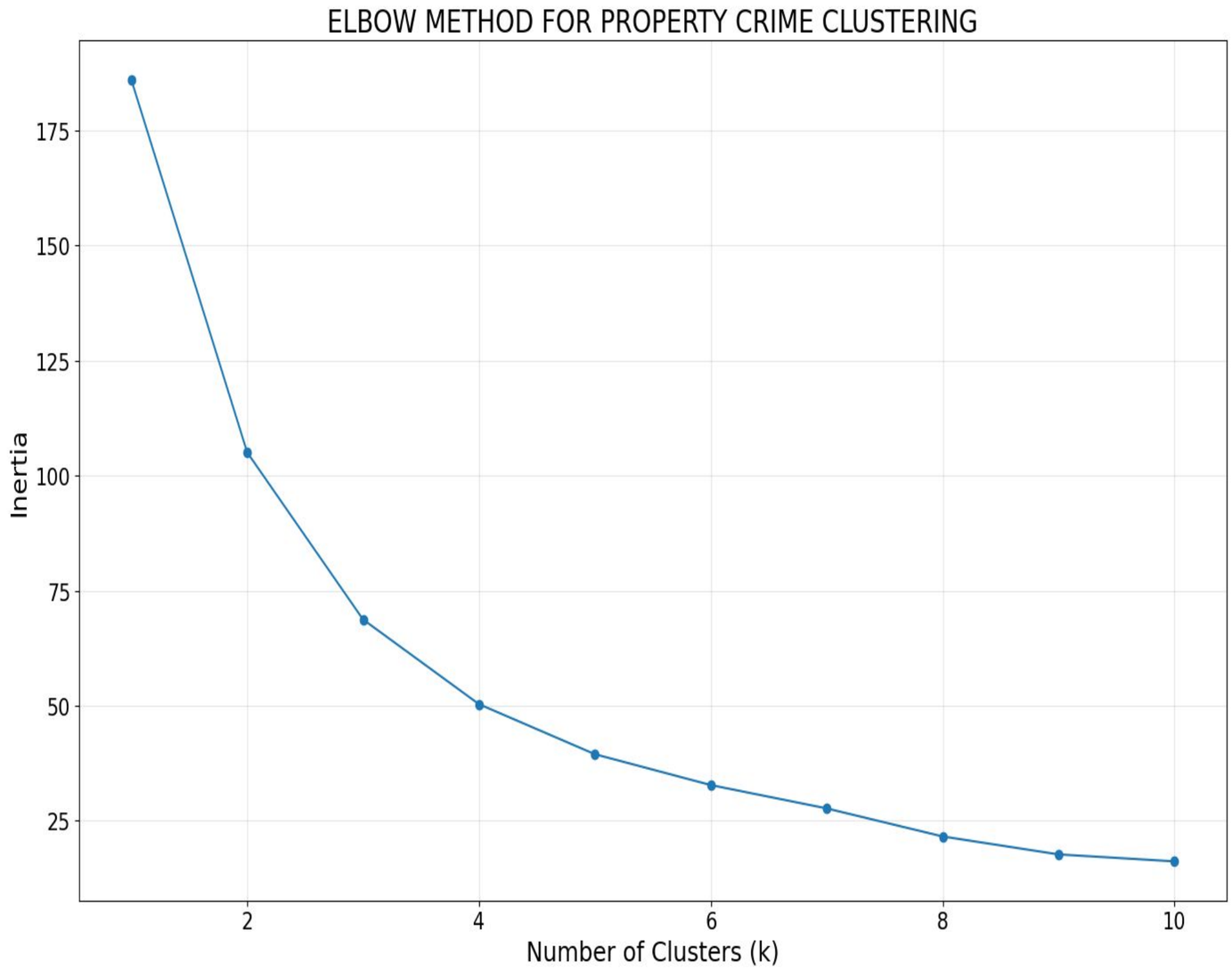
CLUSTERING

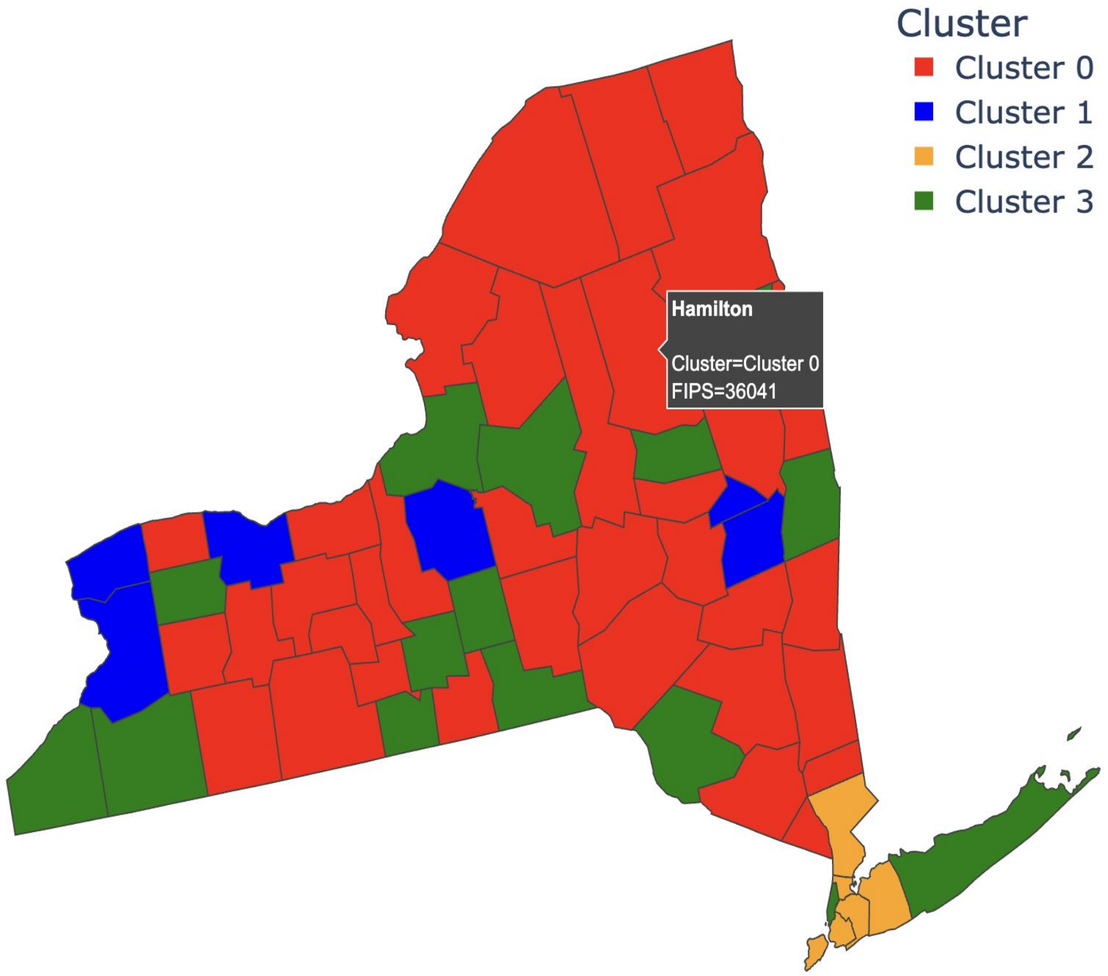


Total variance by PC1 and PC2: 0.9112

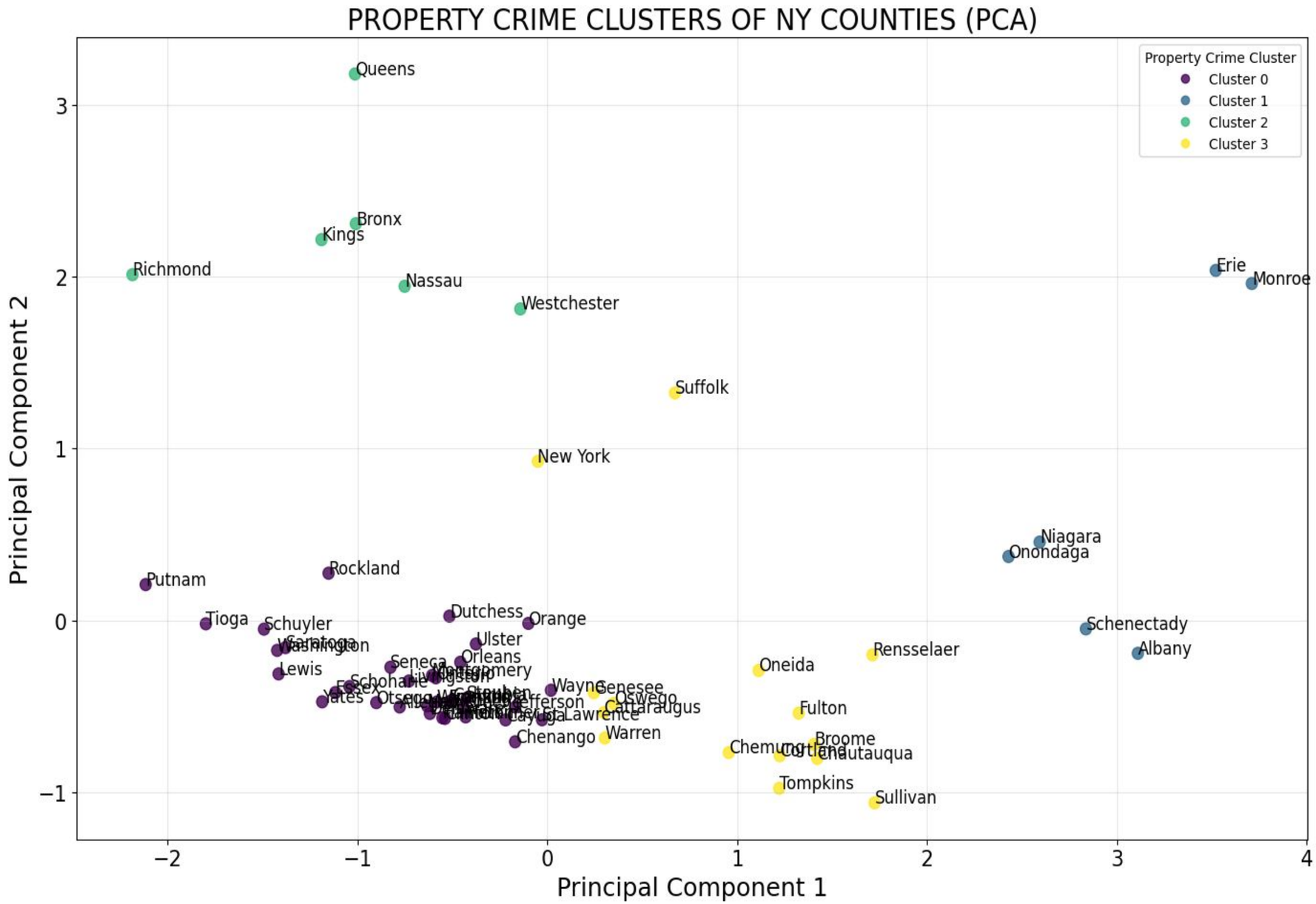
Variable	PC1	PC2
Murder Rate	0.5871	0.106
Rape Rate	-0.2088	0.8932
Robbery Rate	0.5955	-0.1429
Aggravated Assault Rate	0.507	0.4129

CLUSTERING


ELBOW METHOD FOR PROPERTY CRIME CLUSTERING
Inertia
175
150
125
100
75
50
25
2
4
6
8
10
Number of Clusters (k)


Cluster
Cluster 0
Cluster 1
Cluster 2
Cluster 3
Hamilton
Cluster=Cluster 0
FIPS=36041

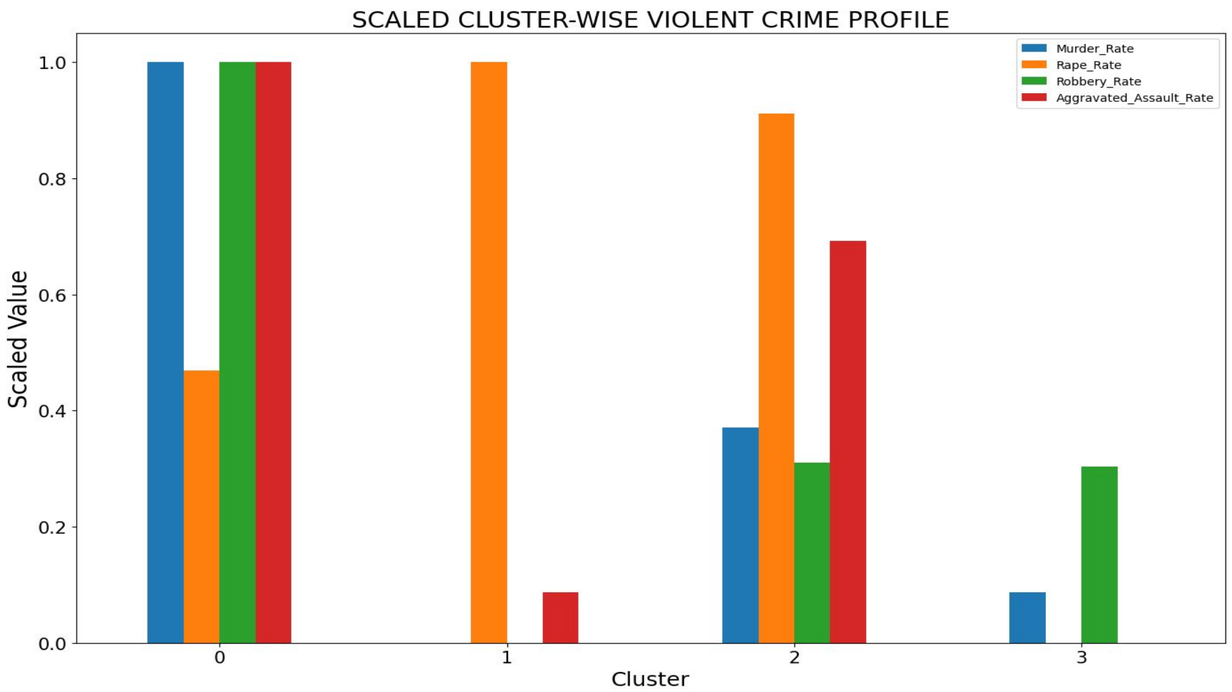
CLUSTERING

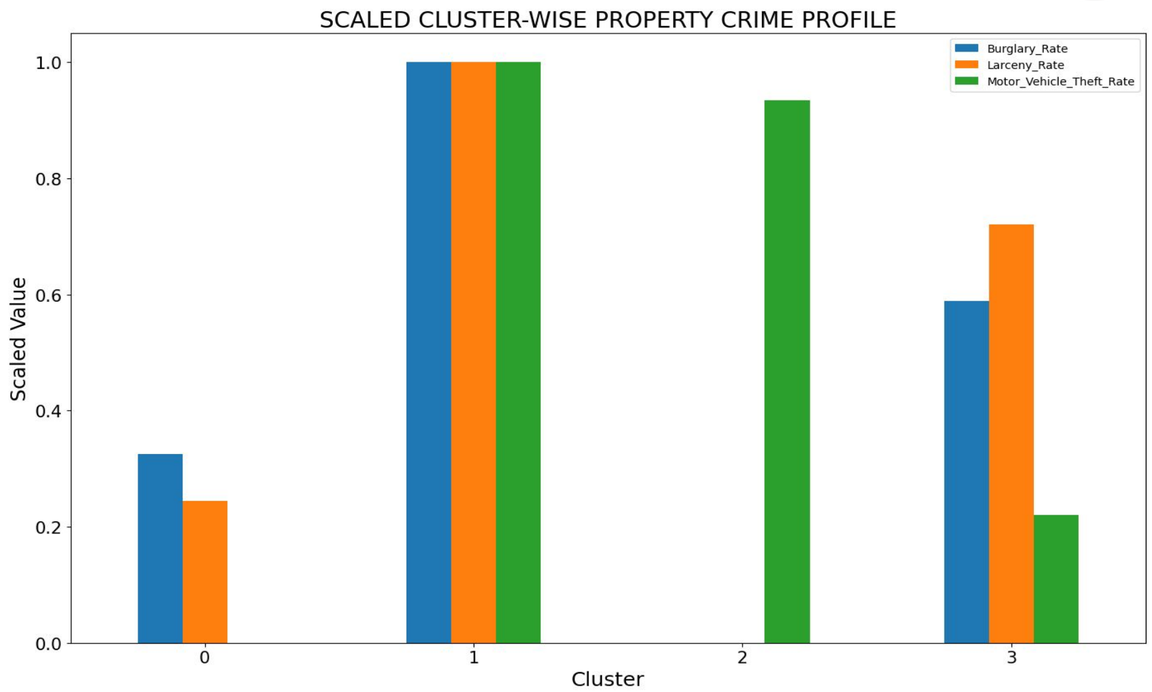


Total variance by PC1 and PC2: 0.89

Variable	PC1	PC2
Burglary Rate	0.6479	-0.3204
Larceny Rate	0.6655	-0.2032
Motor Vehicle Theft Rate	0.3705	0.9252

CRIME PROFILE

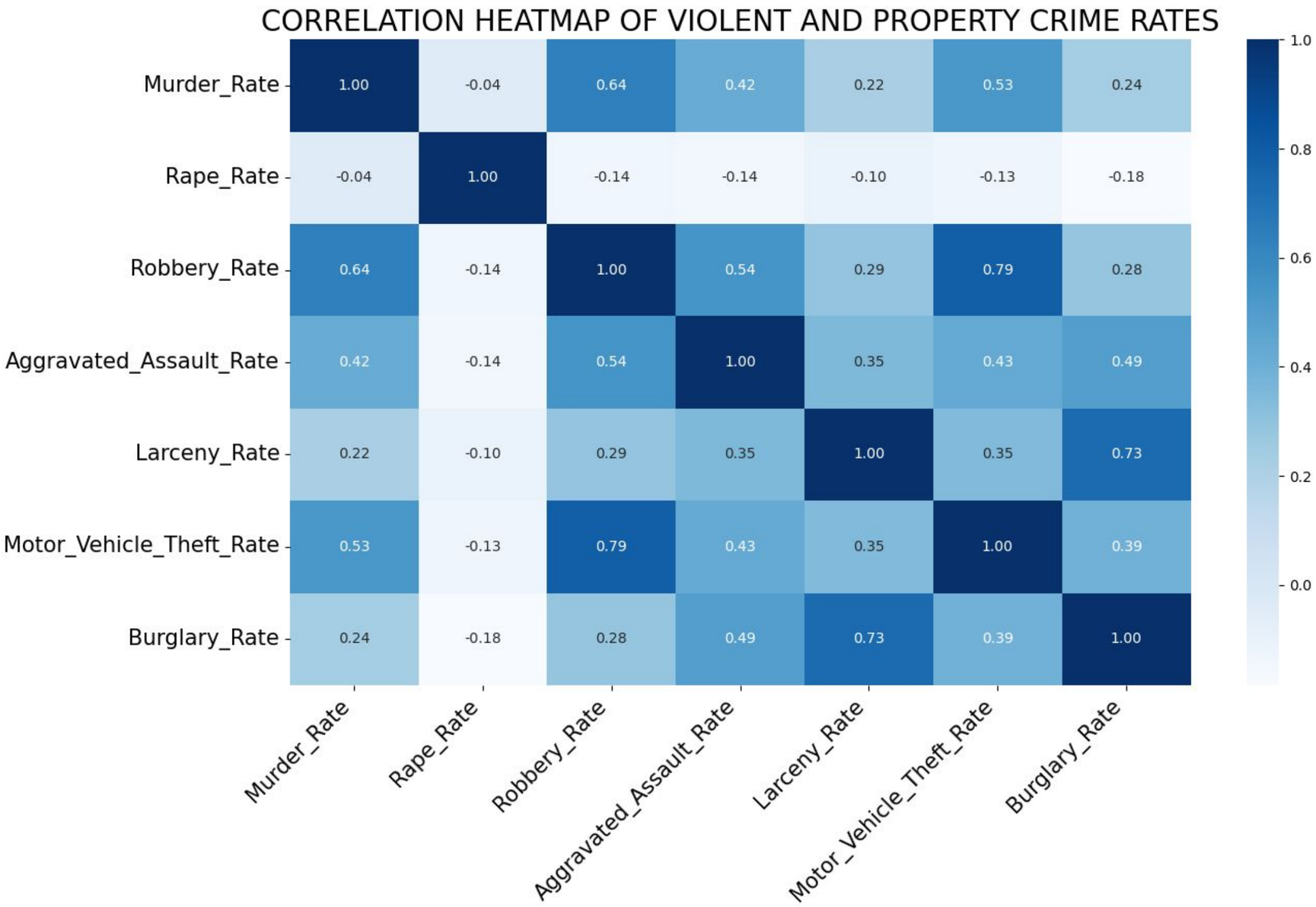




Violent Cluster	Murder Rate	Rape Rate	Robbery Rate	Aggravated Assault Rate
0	12.14	38.99	439.3	523.88
1	3.05	62.7	31.43	257.53
2	6.42	58.76	157.86	434.31
3	3.84	17.98	155.34	232.29

Property Cluster	Burglary Rate	Larceny Rate	Motor Vehicle Theft Rate
0	716.45	2391.26	100.85
1	1270.49	4500.70	481.47
2	449..99	1706.96	456.51
3	933.65	3720.54	184.58

CORRELATION AND MULTICOLLINEARITY ANALYSIS



VIF Before Feature Removal

Feature	VIF
Rape_Rate	2.07
Robbery_Rate	4.32
Aggravated_Assault_Rate	5.62
Larceny_Rate	10.11
Motor_Vehicle_Theft_Rate	4.56
Burglary_Rate	10.18

VIF After Feature Removal

Feature	VIF
Rape_Rate	2.00
Robbery_Rate	4.08
Aggravated_Assault_Rate	4.78
Larceny_Rate	4.70
Motor_Vehicle_Theft_Rate	4.30

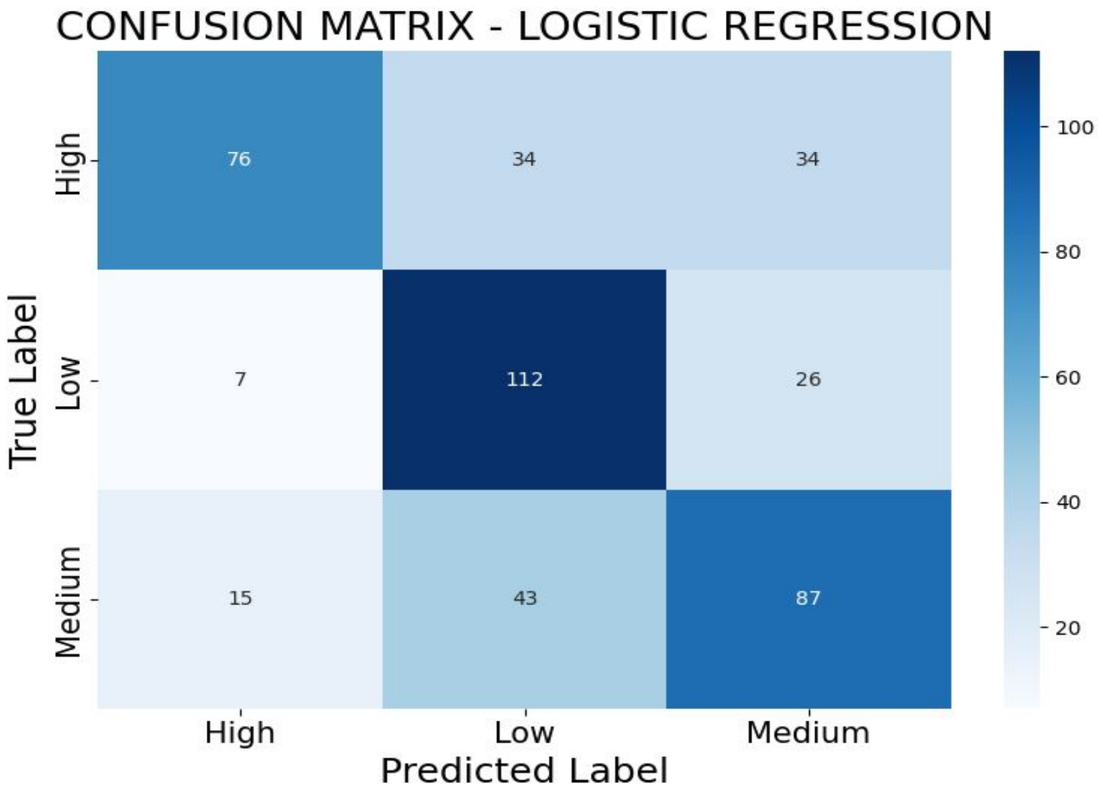
LOGISTIC REGRESSION

Accuracy: 0.63

	Precision	Recall	f1-Score	support
High	0.78	0.53	0.63	144
Low	0.59	0.77	0.67	145
Medium	0.59	0.6	0.6	145

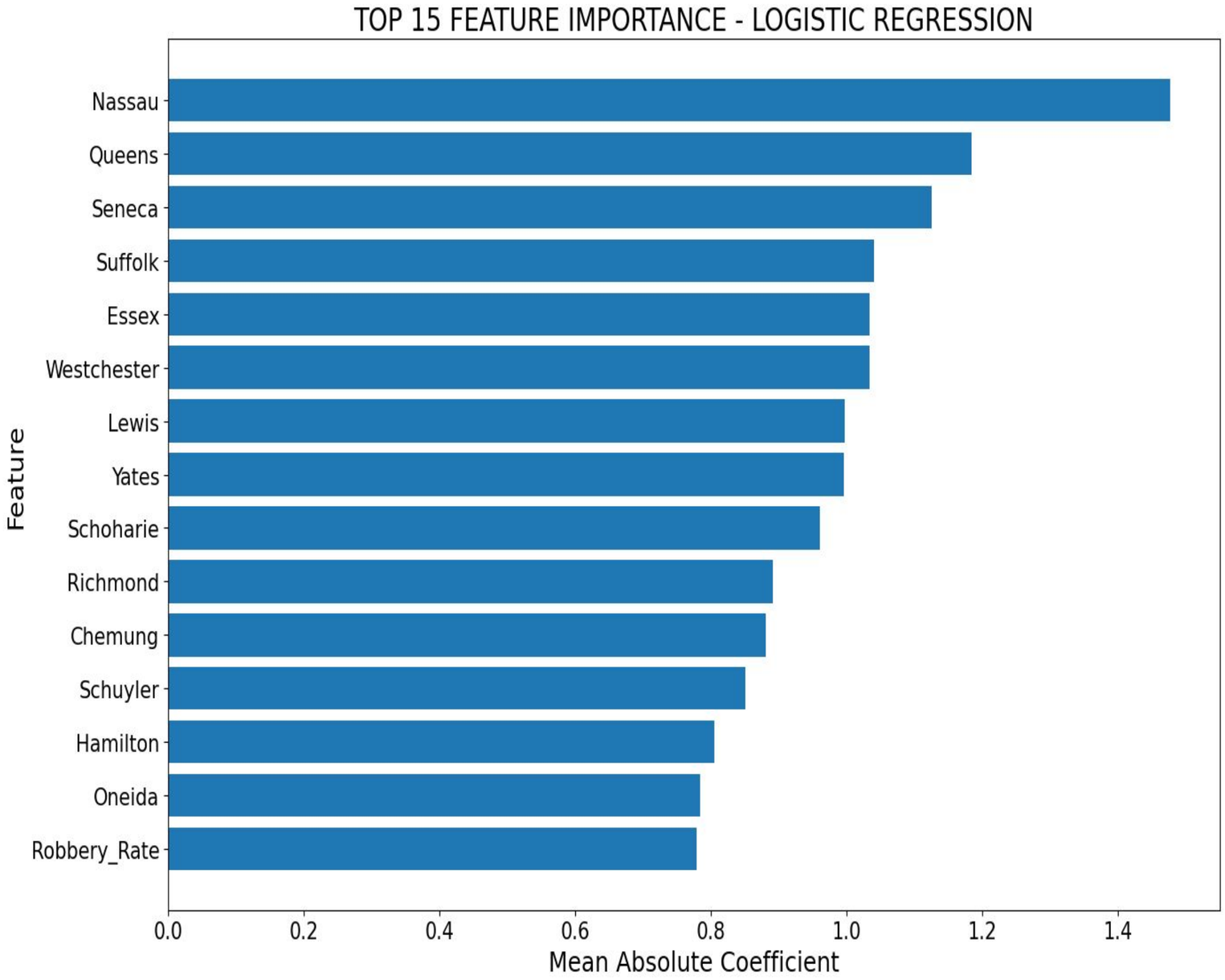
Best model: **C = 1**
Overall performance is **moderate**

logit(P(High)) = 0.19 - 1.89*Nassau + 1.22*Seneca +....
logit(P(Medium)) = 0.06 + 2.21*Nassau + 1.77*Queens+....
logit(P(Low)) = -0.25 - 1.32*Chemung-1.17*Robbery_Rate+....



- Correct predictions are strongest for **Low (112)**
- **High** was often misclassified as **Low** or **Medium**
- **Medium** was frequently confused with **Low**
- The model shows **overlap between classes**

LOGISTIC REGRESSION

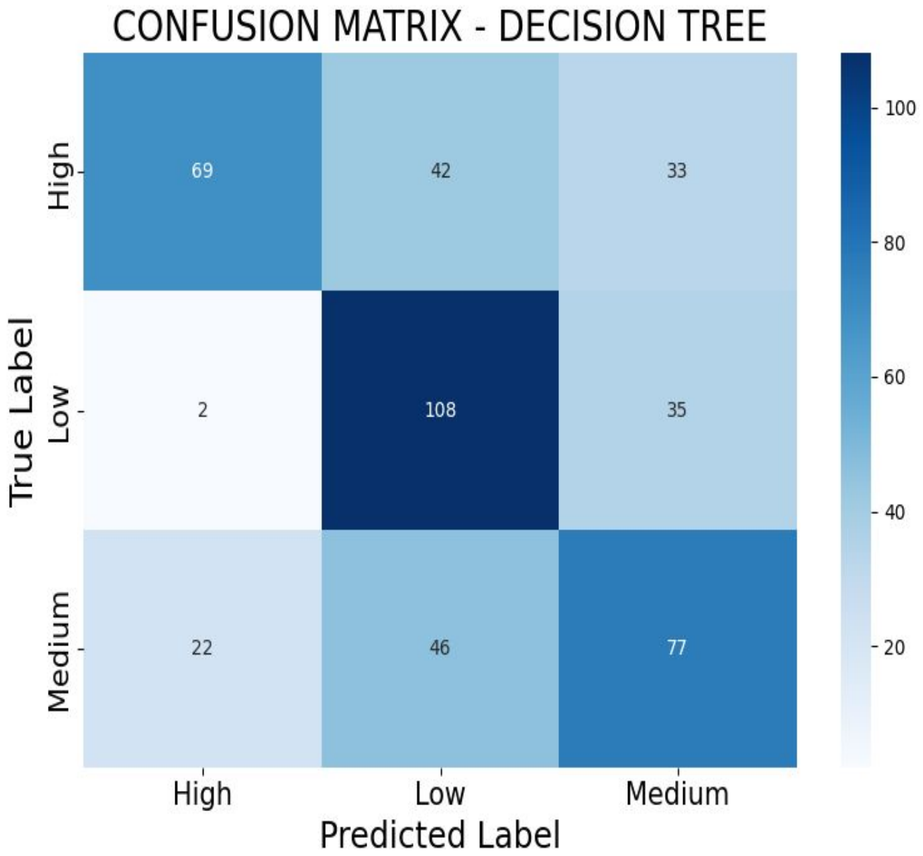


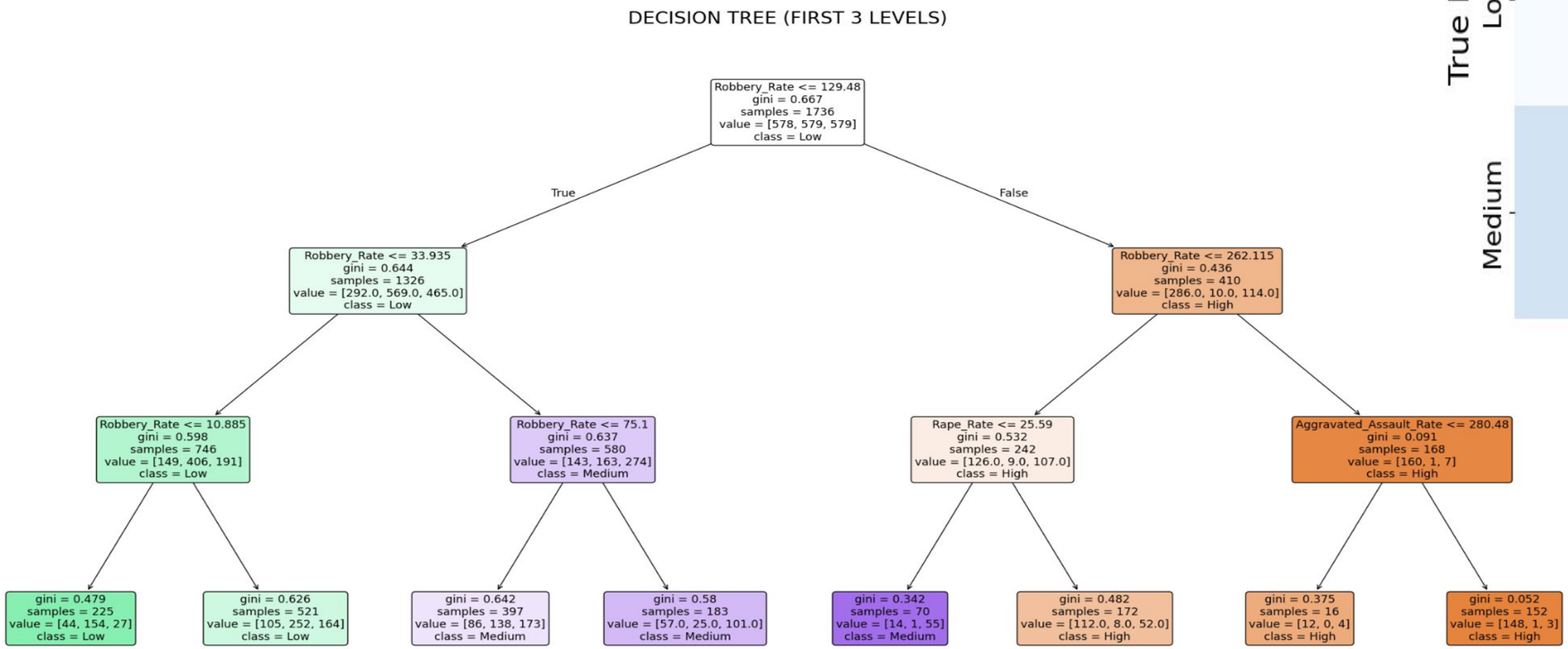
Feature	Coefficient Value
Nassau	1.47
Queens	1.18
Seneca	1.12
Suffolk	1.04
Essex	1.03
Westchester	1.03
Lewis	0.99
Yates	0.99
Schoharie	0.96
Richmond	0.89
Chemung	0.88
Schuyler	0.85
Hamilton	0.80
Oneida	0.78
Robbery_Rate	0.78

DECISION TREE

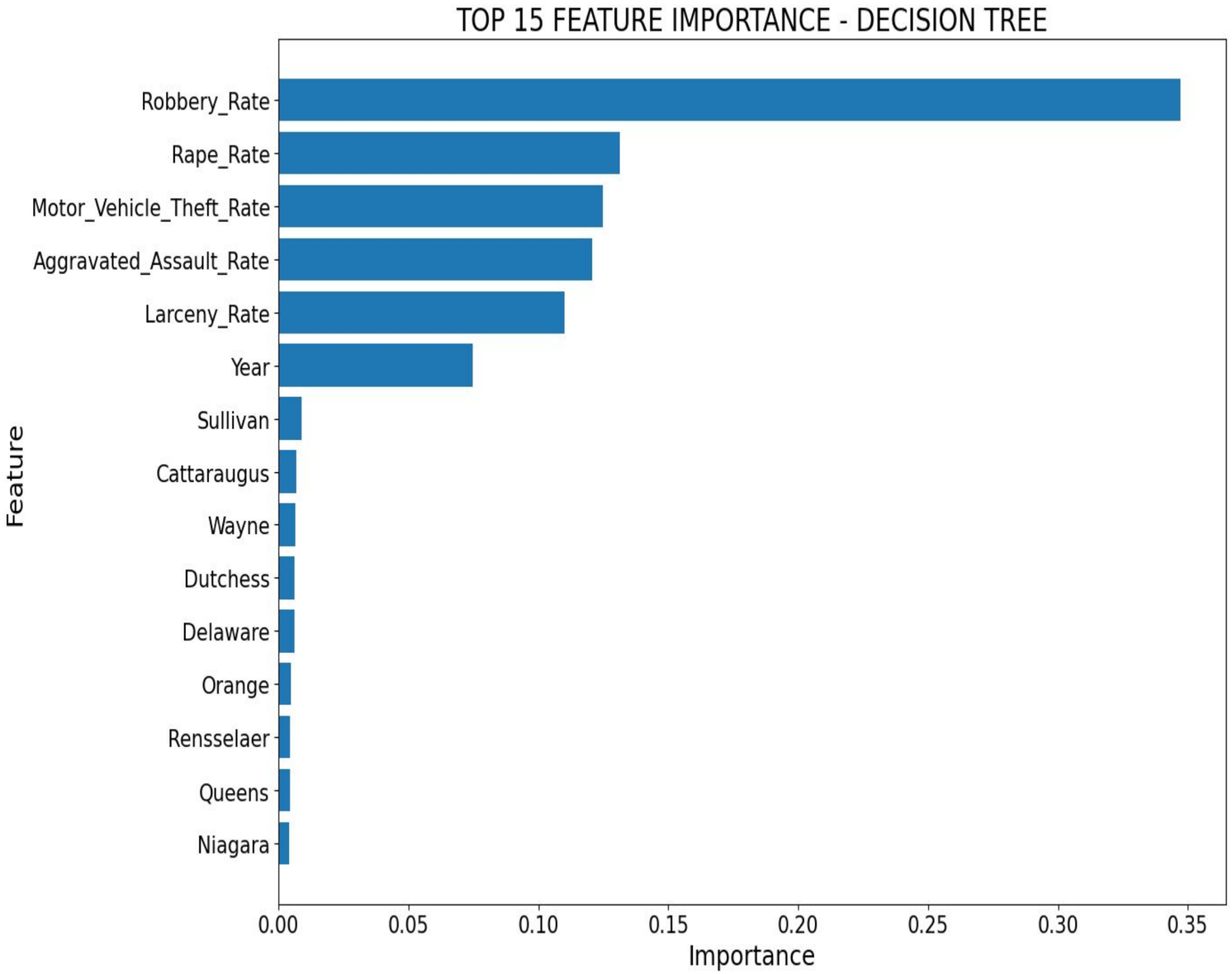
Accuracy: 0.58

Class	Precision	Recall	F1-Score	Support
High	0.74	0.48	0.58	144
Low	0.55	0.74	0.63	145
Medium	0.53	0.53	0.53	145





DECISION TREE

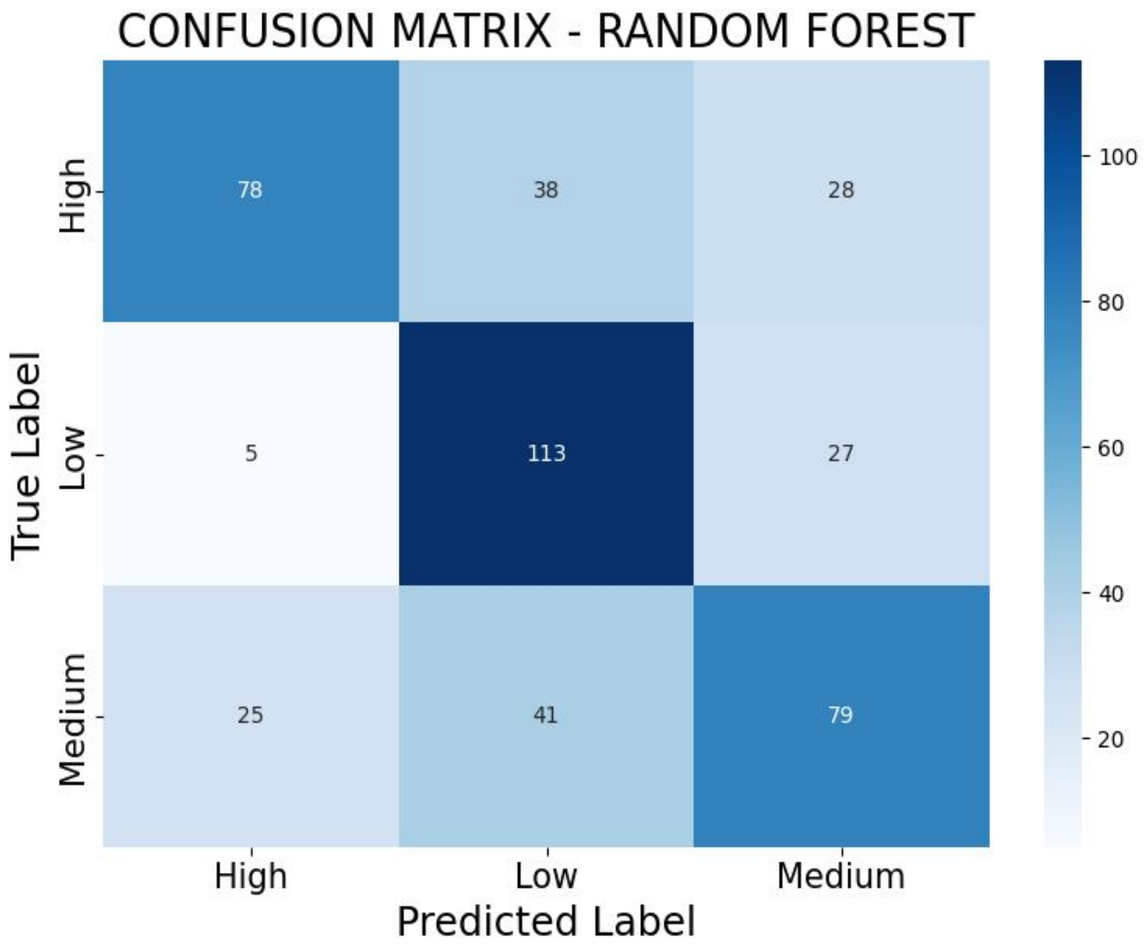


Feature	Importance
Robbery_Rate	0.347294
Rape_Rate	0.131246
Motor_Vehicle_Theft_Rate	0.12471
Aggravated_Assault_Rate	0.120642
Larceny_Rate	0.109949
Year	0.074728
Sullivan	0.008897
Cattaraugus	0.006908
Wayne	0.006572
Dutchess	0.006163
Delaware	0.006155
Orange	0.00456
Rensselaer	0.00433
Queens	0.004207
Niagara	0.004151

RANDOM FOREST

Accuracy: 0.62

Class	Precision	Recall	F1-Score	Support
High	0.72	0.54	0.62	144
Low	0.59	0.78	0.67	145
Medium	0.59	0.54	0.57	145

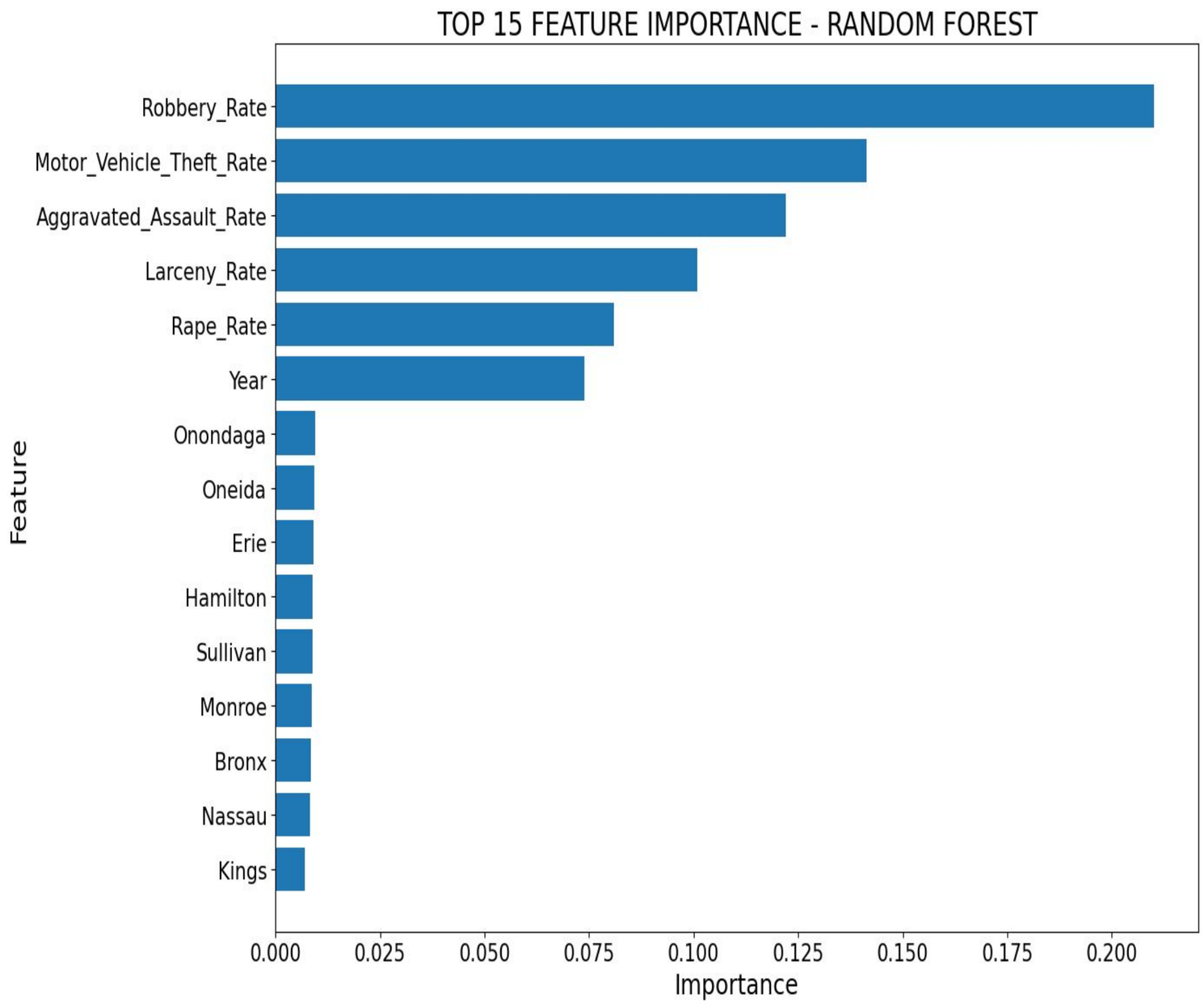


Best model parameters: 100 trees,

Classifier max features: log2

- **Low** was the easiest class to predict.
- **High** was often confused with **Low**.
- **Medium** overlapped with both **High** and **Low**.

RANDOM FOREST



Feature	Importance
Robbery_Rate	0.21
Motor_Vehicle_Theft_Rate	0.14
Aggravated_Assault_Rate	0.12
Larceny_Rate	0.10
Rape_Rate	0.08
Year	0.07
Onondaga	0.009
Oneida	0.009
Erie	0.009
Hamilton	0.008
Sullivan	0.008
Monroe	0.008
Bronx	0.008
Nassau	0.008
Kings	0.007

CONCLUSION

- Cleaned, merged, and standardized crime data using county population.
- Identified clear county-level differences in violent and property crime patterns.
- K-means clustering revealed 4 meaningful groups for both violent and property crime.
- VIF analysis improved feature selection by reducing multicollinearity.
- Logistic Regression performed slightly better than Random Forest..
- Overall, the study showed that data mining methods are effective for understanding and profiling crime across New York counties.

Model	**High**			**Medium**			**Low**			**Accuracy**
	Prec.	Rec.	F1	Prec.	Rec.	F1	Prec.	Rec.	F1	
Logistic Regression	0.78	0.53	0.63	0.59	0.6	0.6	0.59	0.77	0.67	0.63
Decision Tree	0.74	0.48	0.58	0.53	0.53	0.53	0.55	0.74	0.63	0.59
Random Forest	0.72	0.54	0.62	0.59	0.54	0.57	0.59	0.78	0.67	0.62

ACKNOWLEDGEMENT

DR. SUMONA MONDAL

NAVEEN RAMACHANDRA REDDY

QUESTIONS/FEEDBACK

THANKYOU